1.10 Quadratics: The Discriminant

Worked Solutions

1 a)

$(-2)^2 - 4 \times 1 \times -6 = 28$

1 b)

$(0)^2 - 4 \times 1 \times -9 = 36$

1 c)

$(1)^2 - 4 \times -1 \times 5 = 21$

1 d)

$(0)^2 - 4 \times -0.5 \times 12 = 24$

1 e)

$(0)^2 - 4 \times 2 \times 0 = 0$

1 f)

$(-3)^2 - 4 \times 2 \times -2 = -7$

1 g)

$(3)^2 - 4 \times -3 \times -1 = -3$

1 h)

$(-k)^2 - 4 \times 1 \times 2 = k^2 - 8$

2 a)

$16 - 4 \times 1 \times 5 = -4 < 0 \therefore$ no roots

2 b)

$1 - 4 \times 1 \times -6 = 25 > 0 \therefore$ 2 roots

2 c)

$4 - 4 \times 2 \times -4 = 36 \therefore$ 2 roots

2 d)

$9 - 4 \times 8 = -23 \therefore$ 0 roots

2 e)

$64 - 4 \times 1 \times 16 = 0 \therefore$ 1 repeated root

2 f)

$25 - 4 \times -2 \times 3 = 49 \therefore$ 2 roots

2 g)

$144 - 4 \times -4 \times -9 = 0 \therefore$ 1 repeated root

3)

$3x^2 - 3x + 1 = 0$

$(-3)^2 - 4 \times 3 \times 1 = 9 - 12 = -3$

So it has no real solutions.

4 a)

Equal roots is $b^2 - 4ac = 0$

$(k)^2 - 4 \times 4 \times 9 = 0$ so $k^2 = 144$

So $k = -12$ or $k = 12$

4 b)

Equal roots is $b^2 - 4ac = 0$

$(k)^2 - 4 \times 3 \times 12 = 0$ so $k^2 = 144$

So $k = -12$ or $k = 12$

4 c)

Equal roots is $b^2 - 4ac = 0$

$(-3)^2 - 4 \times 1 \times (2 + \frac{1}{k}) = 0$

$9 - 8 - \frac{4}{k} = 0 \Rightarrow 1 = \frac{4}{k}$ so $k = 4$

4 d)

Equal roots is $b^2 - 4ac = 0$

$(2k + 10)^2 - 4 \times 1 \times (k^2 + 5) = 0$

$4k^2 + 40k + 100 - 4(k^2 + 5) = 0$

$40k + 80 = 0$ so $k = -2$

4 e)

Equal roots is $b^2 - 4ac = 0$

$(4)^2 - 4 \times 16 \times k^2 = 0$ so $k^2 = \frac{1}{4}$

So $k = \frac{1}{2}$ or $k = -\frac{1}{2}$

5)

Equal roots is $b^2 - 4ac = 0$

$(3p)^2 - 4 \times 1 \times p = 0$ so $9p^2 - 4p = 0$

$\Rightarrow p(9p - 4) = 0$ so $p = 0$ or $p = \frac{4}{9}$

But p is non-zero so $p = \frac{4}{9}$

6)

$x^2 - (p+q)x + pq$

$(-(p+q))^2 - 4 \times 1 \times pq$

$= (p+q)^2 - 4pq = p^2 + 2pq + q^2 - 4pq$

$= p^2 - 2pq + q^2 = (p-q)^2$

7)

$kx^2 - 2pkx + kp^2 + q$

$(-2pk)^2 - 4 \times k \times (kp^2 + q)$

$= 4p^2k^2 - 4k^2p^2 - 4kq = -4kq$

8)

$y = \frac{5}{2} + x$ so $-\frac{1}{2}x^2 + kx + k = \frac{5}{2} + x$

Giving $-\frac{1}{2}x^2 + (k-1)x + (k - \frac{5}{2}) = 0$

Equal roots is $b^2 - 4ac = 0$:

$(k-1)^2 - 4 \times (-\frac{1}{2})(k - \frac{5}{2}) = 0$

$k^2 - 2k + 1 + 2k - 5 = 0$

$k^2 - 4 = 0$ so $k = -2$ or $k = 2$

9 a)

$y = -\frac{1}{5}x + \frac{1}{5}k$ which gives

$2x^2 + 2(-\frac{1}{5}x + \frac{1}{5}k) = 1$

$10x^2 - 2x + (2k - 5) = 0$

Touches once: $b^2 - 4ac = 0$

so $(-2)^2 - 4 \times 10(2k - 5) = 0$

$4 + 80k + 200 = 0$ so $k = \frac{51}{20}$

9 b)

So the line is $5y + x = \frac{51}{20}$

When $x = 0, y = \frac{51}{100}$ and when $y = 0, x = \frac{51}{20}$

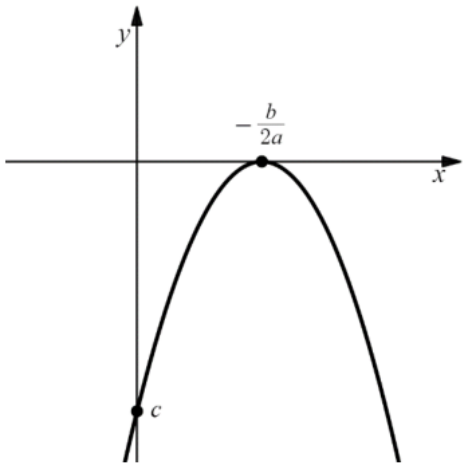
$(0, \frac{51}{100})$ and $(\frac{51}{20}, 0)$

10)

$b^2 - 4ac = 0$, $a < 0$ so negative quadratic

When $x = 0, y = c$

Root at $x = \frac{-b}{2a}$ since $a < 0$ and $b > 0$ the root is positive



11)

For two real roots: $b^2 - 4ac \geq 0$

$y = x^2 + (k-1)x$

$\Rightarrow b^2 - 4ac = (k-1)^2$ which is always ≥ 0

12 a)

$y = 1 - 2x$ gives $x^2 - 4k(1-2x) + 5k = 0$

$x^2 - 4k + 8kx + 5k = 0 \Rightarrow x^2 + 8kx + k = 0$ as req

12 b)

Equal roots is $b^2 - 4ac = 0$

$(8k)^2 - 4 \times 1 \times k = 0$ so $64k^2 - 4k = 0$

$\Rightarrow 4k(16k-1) = 0$ so $k = 0$ or $k = \frac{1}{16}$

Since $k \neq 0, k = \frac{1}{16}$

12 c)

$x^2 - \frac{1}{4}(1-2x) + \frac{5}{16} = 0$

$16x^2 - 4(1-2x) + 5 = 0 \Rightarrow 16x^2 - 4 + 8x + 5 = 0$

$16x^2 + 8x + 1 = 0 \Rightarrow (4x+1)^2 = 0$

$x = -\frac{1}{4}$ and $y = 1 - 2 \times -\frac{1}{4} = \frac{3}{2}$

$(-\frac{1}{4}, \frac{3}{2})$

13)

For equal roots: $b^2 - 4ac = 0$

$(p+q)^2 - 4pq = 0 \Rightarrow p^2 + 2pq + q^2 - 4pq = 0$

$\Rightarrow p^2 - 2pq + q^2 = 0 \Rightarrow (p-q)^2 = 0$ so that $p = q$

14 a)

$x^2 - kx - 4 = -x^2 + x + 4k$

$2x^2 - (k+1)x - (4+4k) = 0$

Touch once: $b^2 - 4ac = 0$

$(-(k+1))^2 - 4 \times 2 \times -(4+4k) = 0$

$k^2 + 2k + 1 + 32 + 32k = 0 \Rightarrow k^2 + 34k + 33 = 0$

$(k+1)(k+33) = 0$ so $k = -1$ or $k = -33$

14 b)

When $k = -1$; $2x^2 = 0$ gives $x = 0$ and $y = -4$

When $k = -33$; $2x^2 + 32x + 128 = 0$

$2(x^2 + 16x + 64) = 0$

$2(x+8)^2 = 0$ gives $x = -8$ and

$y = (-8)^2 - 33 \times 8 - 4 = -204$

$\therefore (0, -4)$ and $(-8, -204)$

15)

Let $p = x^2$

$p^2 - 6p + k = 0$

Equal roots is $b^2 - 4ac = 0$

$36 - 4 \times 1 \times k = 0$ so $36 - 4k = 0$

$\Rightarrow k = 9$